

ASCII and Unicode

Turning text into binary code that can be stored on computers is useful. However, there needs to be agreement on what character each binary number represents.

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◀ 82–83 Binary code

Encoding images

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What is ASCII?

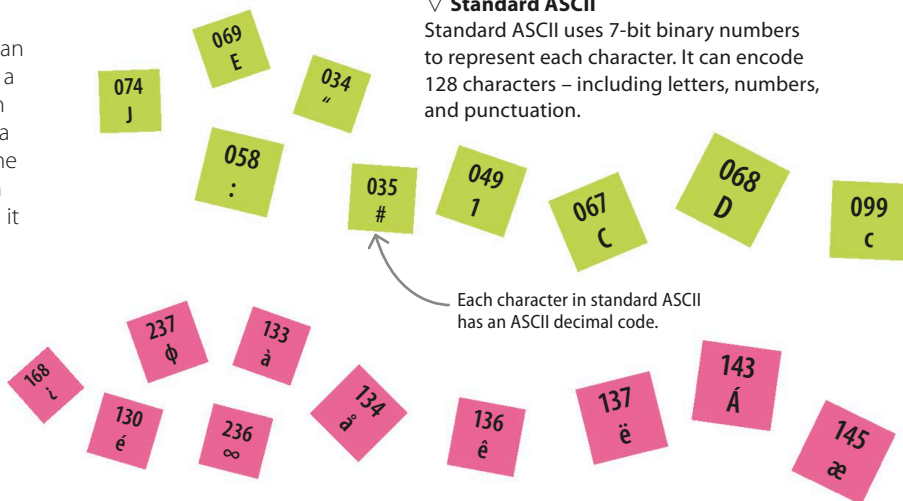
First introduced in 1963, ASCII stands for American Standard Code for Information Interchange. It is a globally agreed standard for representing text in binary code. The standard was developed from a code used by telegraph operators. ASCII became the official standard for computers in the USA in 1968, and many other countries were also using it by the 1980s.

▷ Extended ASCII

Introduced in 1986, extended ASCII uses 8-bit binary numbers. This enables it to encode 256 characters, including letters with accents, useful for a wider variety of languages.

▽ Standard ASCII

Standard ASCII uses 7-bit binary numbers to represent each character. It can encode 128 characters – including letters, numbers, and punctuation.



ASCII files

Most text files are encoded using ASCII. This means that they can be opened, read, and changed using a text editor or a word processor. The files containing code for most computer programs are also encoded using ASCII. This makes it easy for programmers to share open-source code on websites such as GitHub.



▷ Image in text editor

Opening an image in a basic editor often results in pages with strange symbols. The editor interprets the file of binary numbers as text and displays it as ASCII characters.

The information in the image appears as ASCII characters in the text editor.

Open with

Delete

Get info

Rename

Copy

Text Editor

```
≈Ä- 4>b-0Lb"òàbFfrHî"šMÂ1,"
/;[£Áç÷ÔÀµÊ&
W'T'MçB%
RÇEQÓ≤ö¿1Ç'∞Q/
Qg"iÊû`r[Mw√TtgÔzG:ÄçOt=A
!È"%M≥ÿÜ
```

IN DEPTH

ASCII art

It is possible to create images using long lines of ASCII characters. Often called ASCII art, these images were popular in the early days of the internet, as computers lacked the processing power to show proper images. These days, programs can turn images into ASCII art in seconds, such as the example below.

